

# ON THE USE OF TRIGONOMETRIC PARALLAXES FOR THE CALIBRATION OF LUMINOSITY SYSTEMS : THEORY

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Trumpler and Weaver (1953) discuss a systematic error which they attribute to the process of selecting stars by a lower limit in parallax values. The systematic error is introduced because, on the average, the observed parallaxes are larger than the true parallaxes. We demonstrate that the same type of systematic effect exists for all stars with observed trigonometric parallaxes. The problem is treated analytically and can be put in a dimensionless form. The size of the systematic error depends only upon the ratio  $\sigma/\pi_0$  where  $\sigma$  is the standard deviation of  $\pi_0$ , the observed parallax. Corrections are obtained which are to be applied to absolute magnitudes derived from trigonometric parallaxes. The corrected absolute magnitudes are the proper values to use in luminosity calibrations.

*Key words:* luminosity calibration — systematic errors — trigonometric parallaxes

## I. Introduction

It is well known that a systematic error can be introduced when trigonometric parallaxes are used to calibrate a luminosity system. Trumpler and Weaver (1953) point out that "when stars are selected by a lower limit in the observed parallax values, we favor stars for which the measured parallax result is too large. The absolute magnitudes calculated with observed parallax values will thus be systematically too large." This systematic effect is produced because of the errors of observation. Stars actually outside the adopted lower limit will be scattered into the sample and stars inside will be scattered out. However there are more stars just outside the boundary than just inside. Thus more stars will be scattered in than will be scattered out, and the true average parallax for the sample of stars will be smaller than the observed average parallax.

It is instructive to examine the effect at the boundary itself. Let the parallax of the lower limit be  $\pi'$ . Then, of the stars observed to have parallax  $\pi'$ , some will have larger true parallaxes and some will have smaller true parallaxes. Because the number of stars per unit interval of parallax varies as  $1/\pi^4$  (assuming that the stars are uniformly distributed in space), more stars will have their true parallax smaller than  $\pi'$ . Thus the average true parallax for stars with observed parallax  $\pi'$  is smaller than  $\pi'$ .

There is nothing special about the lower par-

allax limit however, and the discussion in the preceding paragraph applies to any value of observed parallax. Thus, when absolute magnitudes obtained from trigonometric parallaxes are used in a luminosity calibration, the possibility of a systematic bias exists. The bias is not caused by the use of a lower parallax limit. Instead it exists at all values of parallax and is a consequence of both the errors of observation and the fact that the number density of stars increases toward smaller parallaxes. We develop a mathematical analysis of the systematic effect in the following sections. A brief review of other investigations of the systematic effect will be given first.

Wallerstein (1967) constructed an artificial sample of stars. He introduced a random error which was added to the true parallax of each star. He then looked at all observed parallaxes within a given limit and compared the absolute magnitudes of the observed sample with those of the true sample inside the same limit. Wallerstein's results suggested that for giant stars, absolute magnitudes obtained from trigonometric parallaxes are too faint on the average by 0.5 to 0.8 magnitude. For stars with  $M_v \geq 5$  he found little or no systematic effect.

Ljunggren and Oja (1965) used an analytical method to obtain corrections which were to be applied to absolute magnitudes obtained from trigonometric parallaxes. These corrections de-

pended upon a number of variables including, among others, the values of observed parallax, dispersion in parallax, and apparent magnitude. We were not aware of the work of Ljunggren and Oja until the present investigation was almost complete.

West (1969) used an analytical method similar to Ljunggren and Oja. He also presents corrections to be applied to observed absolute magnitudes. These are in the form of two-parameter tables for groups of stars meeting certain specific conditions.

In the present work we initially follow the method used by West. We show that the problem can be put in a dimensionless form. An absolute magnitude correction is obtained which depends only upon the ratio  $\sigma/\pi_0$ , where  $\sigma$  is the standard deviation of  $\pi_0$ , the observed parallax.

## II. The Distribution Function

We will assume that the errors of measurement of a trigonometric parallax are such that the distribution of observed parallaxes about the true parallax can be expressed by a Gaussian distribution. Thus the distribution of  $\pi_0$ , the observed parallax, about  $\pi$ , the true parallax, is

$$g(\pi_0 | \pi) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left( -\frac{(\pi_0 - \pi)^2}{2\sigma^2} \right).$$

Here  $\pi = 3.14 \dots$  and  $\sigma$  is the standard deviation.

Conversely, if we have a substantial number of stars with observed parallax  $\pi_0$ , the distribution of true parallaxes about  $\pi_0$  is of interest. If the number of stars with true parallax between  $\pi$  and  $\pi + d\pi$  is constant for all  $\pi$  then it can be shown that the distribution is identical with the one given above. However, this is not the case. If stars are uniformly distributed in space, then the number of them at distances between  $r$  and  $r + dr$  is given by

$$N(r)dr = 4\pi r^2 dr.$$

The number between  $\pi$  and  $\pi + d\pi$  is

$$N(\pi)d\pi = \frac{4\pi d\pi}{\pi^4}.$$

When we take this into account, we find that the distribution of  $\pi$  about  $\pi_0$  can be expressed by

$$g(\pi | \pi_0) \propto \frac{1}{\pi^4} \exp \left( -\frac{(\pi - \pi_0)^2}{2\sigma^2} \right).$$

There are formal difficulties with the constant of proportionality which are discussed in the Appendix. They do not concern us here.

In order to obtain the most probable values of parameters involving the true parallax, we will need to take averages using the distribution function  $g(\pi | \pi_0)$ . However, because of the  $\pi^4$  term in the denominator,  $g(\pi | \pi_0)$  cannot be normalized. Since multiplication by a constant does not alter the proportionality, we write.

$$g(\pi | \pi_0) \propto \left( \frac{\pi_0}{\pi} \right)^4 \exp \left( -\frac{(\pi - \pi_0)^2}{2\sigma^2} \right).$$

The expression on the right is "normalized" now in the sense that it has the value 1 at  $\pi = \pi_0$ , regardless of the value of  $\sigma$ . The new form of the distribution suggests use of a dimensionless parallax  $Z$  defined by

$$Z \equiv \frac{\pi}{\pi_0}.$$

Then

$$g(\pi | \pi_0) \propto G(Z) \equiv \frac{1}{Z^4} \exp \left( -\frac{(Z - 1)^2}{2(\sigma/\pi_0)^2} \right).$$

We have  $G(Z) = 1$  for  $Z = 1$  regardless of the value of  $\sigma/\pi_0$ . Provided  $\sigma$  is fixed, this represents in dimensionless form the distribution of true parallaxes for a particular value of observed parallax. Figure 1 shows a family of curves of this distribution for three values of the parameter  $\sigma/\pi_0$ .

Let us examine the family of curves in Figure 1. Consider first the curve with  $\sigma/\pi_0 = 0.05$ . The distribution of true parallaxes resembles a Gaussian, but it is biased slightly so that its central value is at a value of  $Z$  less than one ( $\pi$  is less than  $\pi_0$ ). This demonstrates the effect of the  $1/Z^4$  term upon the errors of observation. For accurate parallaxes (i.e.  $\sigma/\pi_0$  small) there is a slight shift of the average parallax toward a smaller value of parallax. A comparison of the  $\sigma/\pi_0 = 0.05$  and  $\sigma/\pi_0 = 0.15$  curves reveals that as the errors of observation increase, the shift of the average value of true parallax becomes larger.

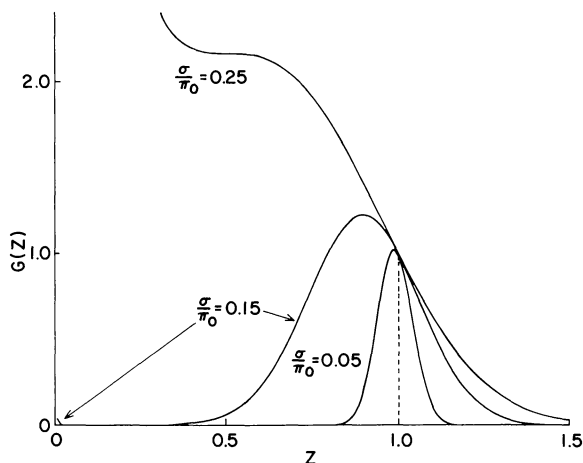


FIG. 1 —  $G(Z)$  vs.  $Z$ . Typical families of curves with parameter  $\sigma/\pi_0$ .

Also, near  $Z = 0.02$  the curve for  $\sigma/\pi_0 = 0.15$  begins to increase. This is due to the divergence of  $1/Z^4$  as  $Z$  approaches zero. The  $\sigma/\pi_0 = 0.05$  curve also diverges in this fashion but it cannot be depicted in Figure 1 because of the scale. This divergence is not physically meaningful. Stars whose parallaxes are in error by such large amounts would be easily recognized. A value of  $Z = 0.02$  implies an error of 11.5 magnitudes in the computed absolute magnitude. Such an error would stand out like a sore thumb. We are only interested in the effect of the  $1/Z^4$  term on the original distribution. The region where  $1/Z^4$  dominates is not relevant and can be ignored.

The third curve shown is the distribution of true parallaxes for the case  $\sigma/\pi_0 = 0.25$ . Because the errors of observation are so large, the  $1/Z^4$  term dominates throughout. There is no possibility of using this distribution to determine the average value of true parallax, because it is not physically realistic. However, the large errors in parallaxes which it predicts are easily recognized when forming luminosity calibrations (see Wilson 1967, Fig. 1 and text). Consequently we will not make gross errors, but also we are not able to correct for the systematic errors which we know must exist.

It is interesting that the size of the error does not depend upon the size of the parallax but rather on the ratio  $\sigma/\pi_0$ . This result is somewhat surprising. Previously it has been thought that the choice of a large parallax for a cutoff value will minimize the systematic effect. This is not

necessarily the case. The critical parameter in determining the size of the error is  $\sigma/\pi_0$ .

In the event that  $\sigma/\pi_0$  is too large, the distribution function is not well behaved (see Fig. 1) and computation of the amount of the error is probably not possible.

We have shown in this section that the distribution of true parallaxes can be expressed in dimensionless form with parameter  $\sigma/\pi_0$ . From Figure 1 it can be seen that we may be able to utilize the distribution function  $G(Z)$  if  $\sigma/\pi_0$  is sufficiently small. For large values of this parameter the task appears to be hopeless. The critical value for  $\sigma/\pi_0$  appears to lie between 0.15 and 0.20.

### III. Estimation of the True Absolute Magnitudes

Let us suppose that we have a large number of stars with observed parallaxes  $\pi_0$ , all having the same value of  $\sigma/\pi_0$ . Then the distribution of their true (dimensionless) parallaxes is given by the  $G(Z)$  corresponding to that particular value of  $\sigma/\pi_0$ . The absolute magnitudes which we would obtain for these stars using  $\pi_0$  are too large on the average because  $G(Z)$  is centered on some value of  $Z$  which is less than one. Since we know  $G(Z)$  we should be able to determine a correction to be applied to these absolute magnitudes. We will use  $G(Z)$  to determine the expected value of  $\Delta M$  where, following West (1969),

$$\Delta M = M_{\text{true}} - M_{\text{observed}} = 5 \log \frac{\pi}{\pi_0} = 5 \log Z.$$

Thus we wish to obtain the expected value of  $5 \log Z$  when the distribution of  $Z$  is given by  $G(Z)$ . The expression for the expected value of  $\Delta M$  is

$$\langle \Delta M \rangle = \frac{5 \int_0^\infty \log Z G(Z) dZ}{\int_0^\infty G(Z) dZ}$$

As written,  $\langle \Delta M \rangle$  is undefined. The integrands in both the numerator and denominator diverge, as  $Z$  approaches zero, in such a way that both integrals are undefined. A modified approach has to be taken. We examine the following expression:

$$\langle \Delta M(\epsilon) \rangle = \frac{5 \int_\epsilon^\infty \log Z G(Z) dZ}{\int_\epsilon^\infty G(Z) dZ}.$$

These integrals have been evaluated numerically for values of  $\epsilon$  ranging from 0.25 to 0.01, and for a range of values of  $\sigma/\pi_0$ . The results are displayed in Figure 2 as a plot of  $\langle \Delta M(\epsilon) \rangle$  versus  $\epsilon$ . The family of curves was generated using  $Z = 1.5$  as the upper limit of integration. Use of a higher value for the upper limit does not affect the results.

Some of the  $\langle \Delta M \rangle$  curves are horizontal for small  $\epsilon$ . The mean value has temporarily stabilized; if we were to continue to extremely small values of  $\epsilon$  we would find that, because of the  $\log Z$  term, these curves also diverge toward  $-\infty$ . The curves which stabilize are those for which a physically meaningful  $G(Z)$  can be obtained. If  $\sigma/\pi_0$  is too large, the distribution of true parallaxes is dominated by the  $1/Z^4$  term, and it is not surprising that useful information about  $\langle \Delta M \rangle$  cannot be obtained. When  $\sigma/\pi_0$  is small, we need not be concerned about what happens as  $Z \rightarrow 0$ . This is because, as discussed in section II, the behavior of  $G(Z)$  for small  $Z$  is irrelevant.

We have listed values of  $\sigma/\pi_0$  and the corresponding stabilized values of  $\langle \Delta M \rangle$  in Table I. These data are also shown in Figure 3. Our results indicate that 0.175 is about the highest value of  $\sigma/\pi_0$  for which the stabilization occurs. Addition of  $\langle \Delta M \rangle$  to the absolute magnitude obtained using  $\pi_0$  will yield a value of absolute

magnitude which is the proper value to use in statistical investigations.

#### IV. Discussion and Conclusions

There are several points of interest. First, it has been assumed by previous investigators that a systematic error is introduced when a lower limit is set on the parallaxes of stars allowed in a sample. We have shown that this error is

TABLE I  
CORRECTIONS TO BE APPLIED  
TO ABSOLUTE MAGNITUDES

| $\sigma/\pi_0$ | $\langle \Delta M \rangle$ |
|----------------|----------------------------|
| 0.0            | 0.0                        |
| 0.025          | -0.01                      |
| 0.050          | -0.02                      |
| 0.075          | -0.06                      |
| 0.100          | -0.11                      |
| 0.125          | -0.18                      |
| 0.150          | -0.28                      |
| 0.175          | -0.43                      |

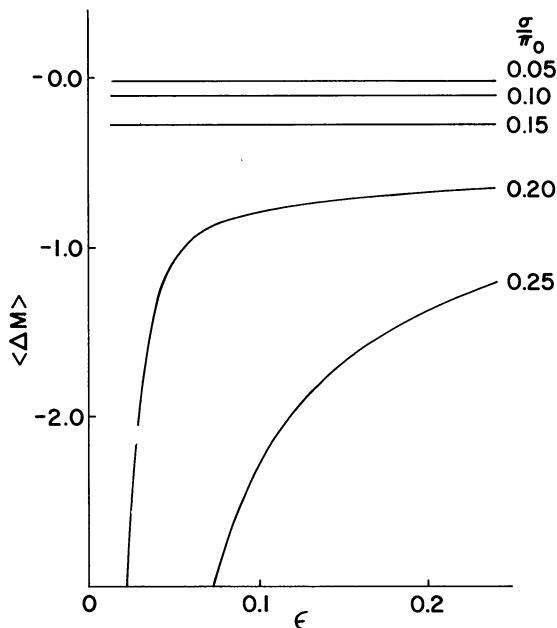


FIG. 2 —  $\langle \Delta M \rangle$  as a function of  $\epsilon$  for various  $\sigma/\pi_0$ .

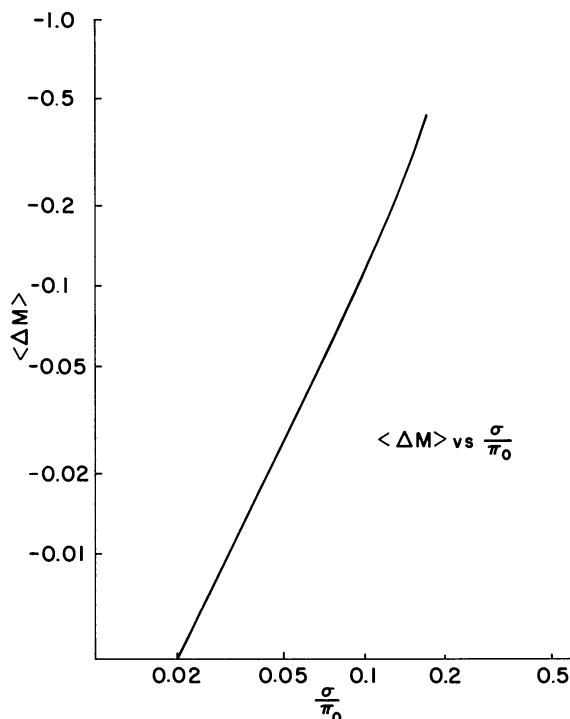


FIG. 3 —  $\langle \Delta M \rangle$  as a function of  $\sigma/\pi_0$ .

present for ALL stars and that it depends upon the ratio  $\sigma/\pi_0$ , not upon the size of  $\pi_0$ . This is a rather surprising result and has serious implications for calibrations based upon trigonometric parallaxes.

A second point is that the allowable values of  $\sigma/\pi_0$  are so highly restrictive. If typical values of  $\sigma$  lie between 0.010 and 0.020 (Vasilevskis 1966, esp. Table III) then  $\pi_0$  needs to be in the range 0.060 to 0.120 or higher in order that we may obtain a correction. This is also a surprising result, since many statistical investigations have been based upon trigonometric parallaxes which are smaller than 0.060 (Wilson 1967; Keenan and Morgan 1951).

Another consideration is related to our assumption of the uniform distribution of stars. This assumption is probably valid out to distances of 50 or 100 parsecs, i.e. the maximum range of accurate parallaxes. Certainly the changes in density within this range are small (McCuskey 1964). Modifications of our work to account for any power-law density distribution are trivial, and one could use the observed density if desirable. However, the assumption of uniform density seems to be a reasonable approximation.

A more interesting point relates to biased parallax samples. Stars are not chosen for parallax measurement at random. If the parallax data are incomplete, a high probability exists that the stars which do have measured parallaxes were selected for some special reason. Depending upon the procedure used for choosing stars whose parallaxes are to be measured, the  $1/Z^4$  term in  $G(Z)$  may have to be modified. We have not yet attempted to investigate this point.

In conclusion, we have shown that a systematic error exists for all trigonometric parallaxes. We have also shown that the distribution of true parallaxes can be put in a dimensionless form. The distribution depends only upon the ratio  $\sigma/\pi_0$ . The distributions have been used to evaluate corrections which are to be applied to absolute magnitudes obtained from observed parallaxes. These corrected absolute magnitudes are the appropriate values to be used in luminosity calibrations.

The results obtained here will be applied to the K-line calibration in a subsequent paper which is being prepared for publication at this time.

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## APPENDIX

In the following we will use the conventional terminology of modern statistics texts (see for example, Lindgren (1962)).

For a given value of  $\pi$ , the conditional density distribution of  $\pi_0$  is

$$f(\pi_0 | \pi = \pi) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(\pi_0 - \pi)^2}{2\sigma^2}\right).$$

It is conventional for density distributions to be normalized. This is not possible for the density distribution of  $\pi$ , the true parallaxes. If we assume a uniform density of stars in space, then

$$f(\pi) \propto \frac{1}{\pi^4}.$$

This function cannot be normalized. Consequently, the joint density distribution of  $\pi$  and  $\pi_0$ , given by

$$f(\pi, \pi_0) = f(\pi)f(\pi_0 | \pi = \pi) \propto \frac{1}{\pi^4} \exp\left(-\frac{(\pi_0 - \pi)^2}{2\sigma^2}\right)$$

cannot be normalized.

In order to obtain the conditional density of  $\pi$  (i.e. the distribution of true parallaxes given a fixed value of  $\pi_0$ , the observed parallax) it is necessary to know the marginal density distribution of  $\pi_0$ . This is given formally by

$$f_{\pi_0}(\pi_0) = \int_0^\infty f(\pi, \pi_0) d\pi.$$

The integral is undefined because of the  $\pi^{-4}$  term in the joint density distribution.

The marginal density distribution enters the conditional density distribution as a normalizing constant in the following manner:

$$f(\pi | \pi_0 = \pi_0) = \frac{f(\pi, \pi_0)}{f_{\pi_0}(\pi_0)} = \frac{f(\pi, \pi_0)}{\int_0^\infty f(\pi, \pi_0) d\pi}.$$

We choose the lower limit of the integration to be very small, but greater than zero. Then  $f_{\Pi_0}(\pi_0)$  is a constant as used here because the conditional density distribution demands that  $\pi_0$  be fixed. Thus  $G(Z)$  defined in section II represents the

conditional density distribution of  $\pi$  to within a multiplicative constant. It is proportional to the distribution of true parallaxes and is even more useful since it is expressed in a dimensionless form.